Bayesian Logistic Regression

Ella Chee

2026-02-27

Bayesian Model for Wildfire Detection

Get data from Kaggle

```
library(kaggler)

# Authenticate creds
#kgl_auth(username = "ellachee21", key = "KGAT_107e527829b7a3f5505e98b50142d7db")

# Download a dataset
# system("/Users/nutellachee/anaconda3/bin/kaggle datasets download -d abdelghaniaaba/wildfire-predicti
```

Load data

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v readr     2.2.0
## v ggplot2   4.0.1     v stringr   1.6.0
## v lubridate 1.9.4     v tibble    3.3.1
## v purrr     1.2.0     v tidyr     1.3.2
```

```
## -- Conflicts ---------------------------------------- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# Helper to load both classes (no wildfire/wildfire) for given folder
load_data <- function(folder_name) {

  nowildfire_list <- list.files(paste0("data/", folder_name, "/nowildfire"), full.names = TRUE)
  wildfire_list <- list.files(paste0("data/", folder_name, "/wildfire"), full.names = TRUE)

  print(sprintf("%d images in no wildfire class", length(nowildfire_list)))
  print(sprintf("%d images in wildfire class", length(wildfire_list)))

  return(list(nowildfire_list, wildfire_list))

}
```

Load validation, training, testing sets

```
# Load validation data
print("VALIDATION:")
```

```
## [1] "VALIDATION:"
```

```
valid_results <- load_data("valid")
```

```
## [1] "2820 images in no wildfire class"
## [1] "3480 images in wildfire class"
```

```
valid_nowildfire <- valid_results[[1]]
valid_wildfire <- valid_results[[2]]

# Load training data
print("TRAINING:")
```

```
## [1] "TRAINING:"
```

```
train_results <- load_data("train")
```

```
## [1] "14499 images in no wildfire class"
## [1] "15750 images in wildfire class"
```

```
train_nowildfire <- train_results[[1]]
train_wildfire <- train_results[[2]]

# Load testing data
print("TESTING:")
```

```
## [1] "TESTING:"
```

```
test_results <- load_data("test")
```

```
## [1] "2820 images in no wildfire class"
## [1] "3479 images in wildfire class"
```

```
test_nowildfire <- test_results[[1]]
test_wildfire <- test_results[[2]]
```

Preview image data

```
library(magick)
```

```
## Linking to ImageMagick 6.9.13.29
## Enabled features: cairo, fontconfig, freetype, heic, lcms, pango, raw, rsvg, webp
## Disabled features: fftw, ghostscript, x11
```

```
# Randomly select image
set.seed(42)
random_img <- sample(test_wildfire, 1)

# Load image into R
img <- image_read(random_img)
print(img)
```

```
## # A tibble: 1 x 7
##   format width height colorspace matte filesize density
##   <chr>  <int>  <int> <chr>      <lgl>    <int> <chr>
## 1 JPEG     350    350 sRGB       FALSE    25099 72x72
```

```
# Load image as array of values
img_array <- as.numeric(image_data(img, channels = "rgb"))
dim(img_array)
```

```
## [1] 350 350   3
```

Data preparation

```
library(magick)
library(dplyr)

# Using each pixel as a separate feature
load_images_to_df <- function(image_paths, label, batch_size = 500, resize = 35) {

  # Batch to increase efficiency
  batches <- split(image_paths, ceiling(seq_along(image_paths) / batch_size))
  results <- lapply(batches, function(batch) {

    imgs <- tryCatch(
      image_read(batch) |>
        image_resize(paste0(resize, "x", resize, "!")),
```

```
      error = function(e) { message("Batch error: ", e$message); NULL }
    )
    if (is.null(imgs)) return(NULL)

    n <- length(batch)
    p <- resize * resize

    # Pre-allocate matrix
    X <- matrix(NA_real_, nrow = n, ncol = p)

    for (i in seq_len(n)) {
      arr <- image_data(imgs[i], channels = "rgb") # [channel, x, y]

      R <- as.integer(arr[1,,]) # Extract separate channels
      G <- as.integer(arr[2,,])
      B <- as.integer(arr[3,,])

      pixel_vals <- R * 256^2 + G * 256 + B # Combine into single value per pixel
      X[i, ] <- as.numeric(pixel_vals) # Flatten to vector
    }
    image_destroy(imgs)

    # Convert to dataframe
    df <- as.data.frame(X)
    colnames(df) <- paste0("px_", seq_len(p))
    #df$name <- batch
    df$label <- label
    df
  })
  bind_rows(results)
}

# Load each dataset
dfs <- list()

for (name in c("train", "valid", "test")) {
  nowildfire_df <- load_images_to_df(get(paste0(name, "_nowildfire")), 0, 500, 16)
  wildfire_df <- load_images_to_df(get(paste0(name, "_wildfire")), 1, 500, 16)

  # Combine and shuffle
  df <- bind_rows(nowildfire_df, wildfire_df)
  set.seed(42)
  shuffled_df <- df[sample(1:nrow(df)), ]

  dfs[[name]] <- shuffled_df
  print(sprintf("Loaded %s dataset", name))
}
```

```
## [1] "Loaded train dataset"
## [1] "Loaded valid dataset"
## [1] "Loaded test dataset"
```

```r
# Apply to each dataset
train_df <- dfs[["train"]]
valid_df <- dfs[["valid"]]
test_df <- dfs[["test"]]
```

Model set-up

1. Resized 350x350px to 35x35px –> flattened to 1225 px values = 1225 features, with chains = 4, iter = 20, warmup = 3, 68 divergent chains after warmup, largest R-hat is Inf, indicating chains have not mixed, vector memory limit of 24.0 GB reached
2. Resized to 16x16 px –> flattened to 256 px values = 256 features, with chains = 4, iter = 100, warmup = 18, 327 divergent transitions after warmup, largest R-hat is 30.33, indicating chains have not mixed vector memory limit of 24.0 Gb reached, see mem.maxVSize()
3. Resized to 16x16 px –> flattened to 256 px values = 256 features, with chains = 4, iter = 500, warmup = 90, 616 divergent transitions after warmup, largest R-hat is R-hat is 7.38, indicating chains have not mixed vector memory limit of 24.0 Gb reached, see mem.maxVSize()
4. Resized to 16x16 px –> flattened to 256 px values = 256 features, with chains = 4, iter = 1000, warmup = 180, 199 divergent transitions after warmup, largest R-hat is R-hat is 4.75, indicating chains have not mixed
5. Resized to 16x16 px –> flattened to 256 px values = 256 features, with chains = 6, iter = 10000, warmup = 1800, 2540 divergent transitions after warmup, largest R-hat is R-hat is 5.48, indicating chains have not mixed

```r
library(brms)
```

```
## Loading required package: Rcpp
```

```
## Loading 'brms' package (version 2.23.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').
```

```
##
## Attaching package: 'brms'
```

```
## The following object is masked from 'package:stats':
##
##     ar
```

```r
# Set default prior (use Bernoulli for binary classification)
default_prior <- default_prior(label ~ .,
              data = train_df,
              family = bernoulli("logit"))

# Creating Bayesian logistic regression model, use all pixels as features
brm_logit <- brm(label ~ .,
                 family = bernoulli("logit"),
                 data = train_df,
                 chains = 6,
```

```
                iter = 10000,
                warmup = 1800,
                prior = default_prior)
```

```
## Compiling Stan program...

## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: 'MacOSX14.2.sdk'
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Li
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/StanHeade
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/
## /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/include/Eigen/src/Cor
## #include <cmath>
##          ^~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1

## Start sampling

##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
## Chain 1:
## Chain 1: Gradient evaluation took 0.002033 seconds
## Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 20.33 seconds.
## Chain 1: Adjust your expectations accordingly!
## Chain 1:
## Chain 1:
## Chain 1: Iteration:    1 / 10000 [  0%]  (Warmup)
## Chain 1: Iteration: 1000 / 10000 [ 10%]  (Warmup)
## Chain 1: Iteration: 1801 / 10000 [ 18%]  (Sampling)
## Chain 1: Iteration: 2800 / 10000 [ 28%]  (Sampling)
## Chain 1: Iteration: 3800 / 10000 [ 38%]  (Sampling)
## Chain 1: Iteration: 4800 / 10000 [ 48%]  (Sampling)
## Chain 1: Iteration: 5800 / 10000 [ 58%]  (Sampling)
## Chain 1: Iteration: 6800 / 10000 [ 68%]  (Sampling)
## Chain 1: Iteration: 7800 / 10000 [ 78%]  (Sampling)
## Chain 1: Iteration: 8800 / 10000 [ 88%]  (Sampling)
## Chain 1: Iteration: 9800 / 10000 [ 98%]  (Sampling)
## Chain 1: Iteration: 10000 / 10000 [100%]  (Sampling)
## Chain 1:
## Chain 1:  Elapsed Time: 33.574 seconds (Warm-up)
## Chain 1:                55.683 seconds (Sampling)
## Chain 1:                89.257 seconds (Total)
## Chain 1:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
## Chain 2:
```

```
## Chain 2: Gradient evaluation took 0.002744 seconds
## Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 27.44 seconds.
## Chain 2: Adjust your expectations accordingly!
## Chain 2:
## Chain 2:
## Chain 2: Iteration:    1 / 10000 [  0%]  (Warmup)
## Chain 2: Iteration: 1000 / 10000 [ 10%]  (Warmup)
## Chain 2: Iteration: 1801 / 10000 [ 18%]  (Sampling)
## Chain 2: Iteration: 2800 / 10000 [ 28%]  (Sampling)
## Chain 2: Iteration: 3800 / 10000 [ 38%]  (Sampling)
## Chain 2: Iteration: 4800 / 10000 [ 48%]  (Sampling)
## Chain 2: Iteration: 5800 / 10000 [ 58%]  (Sampling)
## Chain 2: Iteration: 6800 / 10000 [ 68%]  (Sampling)
## Chain 2: Iteration: 7800 / 10000 [ 78%]  (Sampling)
## Chain 2: Iteration: 8800 / 10000 [ 88%]  (Sampling)
## Chain 2: Iteration: 9800 / 10000 [ 98%]  (Sampling)
## Chain 2: Iteration: 10000 / 10000 [100%]  (Sampling)
## Chain 2:
## Chain 2:  Elapsed Time: 34.827 seconds (Warm-up)
## Chain 2:                75.129 seconds (Sampling)
## Chain 2:                109.956 seconds (Total)
## Chain 2:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
## Chain 3:
## Chain 3: Gradient evaluation took 0.001952 seconds
## Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 19.52 seconds.
## Chain 3: Adjust your expectations accordingly!
## Chain 3:
## Chain 3:
## Chain 3: Iteration:    1 / 10000 [  0%]  (Warmup)
## Chain 3: Iteration: 1000 / 10000 [ 10%]  (Warmup)
## Chain 3: Iteration: 1801 / 10000 [ 18%]  (Sampling)
## Chain 3: Iteration: 2800 / 10000 [ 28%]  (Sampling)
## Chain 3: Iteration: 3800 / 10000 [ 38%]  (Sampling)
## Chain 3: Iteration: 4800 / 10000 [ 48%]  (Sampling)
## Chain 3: Iteration: 5800 / 10000 [ 58%]  (Sampling)
## Chain 3: Iteration: 6800 / 10000 [ 68%]  (Sampling)
## Chain 3: Iteration: 7800 / 10000 [ 78%]  (Sampling)
## Chain 3: Iteration: 8800 / 10000 [ 88%]  (Sampling)
## Chain 3: Iteration: 9800 / 10000 [ 98%]  (Sampling)
## Chain 3: Iteration: 10000 / 10000 [100%]  (Sampling)
## Chain 3:
## Chain 3:  Elapsed Time: 33.475 seconds (Warm-up)
## Chain 3:                51.405 seconds (Sampling)
## Chain 3:                84.88 seconds (Total)
## Chain 3:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
## Chain 4:
## Chain 4: Gradient evaluation took 0.001997 seconds
## Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 19.97 seconds.
## Chain 4: Adjust your expectations accordingly!
## Chain 4:
```

```
## Chain 4:
## Chain 4: Iteration:    1 / 10000 [  0%]  (Warmup)
## Chain 4: Iteration: 1000 / 10000 [ 10%]  (Warmup)
## Chain 4: Iteration: 1801 / 10000 [ 18%]  (Sampling)
## Chain 4: Iteration: 2800 / 10000 [ 28%]  (Sampling)
## Chain 4: Iteration: 3800 / 10000 [ 38%]  (Sampling)
## Chain 4: Iteration: 4800 / 10000 [ 48%]  (Sampling)
## Chain 4: Iteration: 5800 / 10000 [ 58%]  (Sampling)
## Chain 4: Iteration: 6800 / 10000 [ 68%]  (Sampling)
## Chain 4: Iteration: 7800 / 10000 [ 78%]  (Sampling)
## Chain 4: Iteration: 8800 / 10000 [ 88%]  (Sampling)
## Chain 4: Iteration: 9800 / 10000 [ 98%]  (Sampling)
## Chain 4: Iteration: 10000 / 10000 [100%]  (Sampling)
## Chain 4:
## Chain 4:  Elapsed Time: 35.516 seconds (Warm-up)
## Chain 4:                157.065 seconds (Sampling)
## Chain 4:                192.581 seconds (Total)
## Chain 4:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 5).
## Chain 5:
## Chain 5: Gradient evaluation took 0.002032 seconds
## Chain 5: 1000 transitions using 10 leapfrog steps per transition would take 20.32 seconds.
## Chain 5: Adjust your expectations accordingly!
## Chain 5:
## Chain 5:
## Chain 5: Iteration:    1 / 10000 [  0%]  (Warmup)
## Chain 5: Iteration: 1000 / 10000 [ 10%]  (Warmup)
## Chain 5: Iteration: 1801 / 10000 [ 18%]  (Sampling)
## Chain 5: Iteration: 2800 / 10000 [ 28%]  (Sampling)
## Chain 5: Iteration: 3800 / 10000 [ 38%]  (Sampling)
## Chain 5: Iteration: 4800 / 10000 [ 48%]  (Sampling)
## Chain 5: Iteration: 5800 / 10000 [ 58%]  (Sampling)
## Chain 5: Iteration: 6800 / 10000 [ 68%]  (Sampling)
## Chain 5: Iteration: 7800 / 10000 [ 78%]  (Sampling)
## Chain 5: Iteration: 8800 / 10000 [ 88%]  (Sampling)
## Chain 5: Iteration: 9800 / 10000 [ 98%]  (Sampling)
## Chain 5: Iteration: 10000 / 10000 [100%]  (Sampling)
## Chain 5:
## Chain 5:  Elapsed Time: 27.772 seconds (Warm-up)
## Chain 5:                45.955 seconds (Sampling)
## Chain 5:                73.727 seconds (Total)
## Chain 5:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 6).
## Chain 6:
## Chain 6: Gradient evaluation took 0.002109 seconds
## Chain 6: 1000 transitions using 10 leapfrog steps per transition would take 21.09 seconds.
## Chain 6: Adjust your expectations accordingly!
## Chain 6:
## Chain 6:
## Chain 6: Iteration:    1 / 10000 [  0%]  (Warmup)
## Chain 6: Iteration: 1000 / 10000 [ 10%]  (Warmup)
## Chain 6: Iteration: 1801 / 10000 [ 18%]  (Sampling)
```

```
## Chain 6: Iteration: 2800 / 10000 [ 28%]  (Sampling)
## Chain 6: Iteration: 3800 / 10000 [ 38%]  (Sampling)
## Chain 6: Iteration: 4800 / 10000 [ 48%]  (Sampling)
## Chain 6: Iteration: 5800 / 10000 [ 58%]  (Sampling)
## Chain 6: Iteration: 6800 / 10000 [ 68%]  (Sampling)
## Chain 6: Iteration: 7800 / 10000 [ 78%]  (Sampling)
## Chain 6: Iteration: 8800 / 10000 [ 88%]  (Sampling)
## Chain 6: Iteration: 9800 / 10000 [ 98%]  (Sampling)
## Chain 6: Iteration: 10000 / 10000 [100%]  (Sampling)
## Chain 6:
## Chain 6:  Elapsed Time: 33.551 seconds (Warm-up)
## Chain 6:                174.594 seconds (Sampling)
## Chain 6:                208.145 seconds (Total)
## Chain 6:

## Warning: There were 20258 divergent transitions after warmup. See
## https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
## to find out why this is a problem and how to eliminate them.

## Warning: There were 6 chains where the estimated Bayesian Fraction of Missing Information was low. S
## https://mc-stan.org/misc/warnings.html#bfmi-low

## Warning: Examine the pairs() plot to diagnose sampling problems

## Warning: The largest R-hat is 7.52, indicating chains have not mixed.
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#r-hat

## Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians may be
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#bulk-ess

## Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tail quant
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#tail-ess
```

```
# Get summary stats on posterior
summary(brm_logit)
```

```
## Warning: Parts of the model have not converged (some Rhats are > 1.05). Be
## careful when analysing the results! We recommend running more iterations and/or
## setting stronger priors.

## Warning: There were 20258 divergent transitions after warmup. Increasing
## adapt_delta above 0.8 may help. See
## http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

##  Family: bernoulli
##   Links: mu = logit
## Formula: label ~ px_1 + px_2 + px_3 + px_4 + px_5 + px_6 + px_7 + px_8 + px_9 + px_10 + px_11 + px_1
##    Data: train_df (Number of observations: 30249)
```

```
##   Draws: 6 chains, each with iter = 10000; warmup = 1800; thin = 1;
##          total post-warmup draws = 49200
##
## Regression Coefficients:
##             Estimate  Est.Error   l-95% CI   u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept 3771443.66 2140362.31 1153468.37 7593039.24 4.79        6        6
## px_1           -0.09       0.06      -0.25      -0.02 5.29        6       11
## px_2           -0.04       0.15      -0.34       0.21 5.67        6        6
## px_3            0.04       0.12      -0.16       0.22 5.62        6       11
## px_4           -0.05       0.12      -0.28       0.13 5.77        6        6
## px_5            0.04       0.11      -0.16       0.22 5.40        6       11
## px_6           -0.07       0.11      -0.24       0.13 5.20        6        6
## px_7            0.07       0.11      -0.07       0.26 5.56        6       11
## px_8           -0.05       0.10      -0.20       0.11 4.89        6        6
## px_9            0.01       0.14      -0.28       0.11 3.76        7        6
## px_10          -0.00       0.10      -0.20       0.18 6.84        6       11
## px_11           0.02       0.07      -0.12       0.15 5.16        6        6
## px_12          -0.09       0.06      -0.24      -0.01 3.34        7       11
## px_13          -0.00       0.08      -0.15       0.14 4.76        6        6
## px_14          -0.04       0.14      -0.29       0.13 5.47        6        6
## px_15           0.04       0.18      -0.18       0.32 3.82        7        6
## px_16          -0.07       0.13      -0.33       0.07 5.29        6        6
## px_17           0.06       0.11      -0.07       0.28 6.16        6        6
## px_18           0.06       0.25      -0.33       0.49 5.29        6       11
## px_19          -0.06       0.20      -0.32       0.26 5.72        6        6
## px_20           0.06       0.20      -0.28       0.39 5.54        6       11
## px_21          -0.07       0.21      -0.37       0.32 5.26        6        6
## px_22           0.07       0.15      -0.22       0.26 5.50        6       11
## px_23          -0.11       0.17      -0.38       0.06 5.23        7        6
## px_24           0.06       0.15      -0.22       0.28 5.30        6        6
## px_25           0.02       0.27      -0.16       0.60 4.24        6       22
## px_26          -0.08       0.23      -0.56       0.15 5.66        6        6
## px_27           0.06       0.18      -0.09       0.45 5.55        6       11
## px_28           0.02       0.02      -0.01       0.09 2.36        7        6
## px_29           0.05       0.11      -0.05       0.25 4.79        6       11
## px_30           0.02       0.12      -0.16       0.22 5.28        6        6
## px_31          -0.02       0.14      -0.23       0.16 4.88        6        6
## px_32           0.01       0.10      -0.15       0.18 6.95        6       NA
## px_33          -0.07       0.07      -0.21       0.02 5.88        6       NA
## px_34          -0.08       0.17      -0.31       0.22 5.07        6        6
## px_35           0.11       0.20      -0.16       0.50 5.02        6       11
## px_36          -0.10       0.23      -0.49       0.25 6.62        6        6
## px_37           0.08       0.23      -0.27       0.45 5.40        6       11
## px_38          -0.00       0.13      -0.20       0.21 7.26        6        6
## px_39           0.02       0.20      -0.19       0.43 5.34        6       11
## px_40          -0.00       0.08      -0.09       0.13 4.93        6       11
## px_41          -0.05       0.19      -0.46       0.10 5.58        6        6
## px_42           0.09       0.17      -0.08       0.44 3.55        7        6
## px_43          -0.10       0.20      -0.51       0.12 5.03        6        6
## px_44           0.01       0.10      -0.18       0.13 4.98        6        6
## px_45          -0.03       0.06      -0.13       0.04 3.39        7        6
## px_46          -0.01       0.07      -0.13       0.08 4.53        6       11
## px_47          -0.01       0.08      -0.13       0.09 4.56        6       NA
## px_48          -0.01       0.09      -0.19       0.11 5.96        6        6
```

```
## px_49          0.06       0.04        0.01       0.15 5.16        6        11
## px_50          0.00       0.09       -0.17       0.15 5.83        6        11
## px_51         -0.02       0.13       -0.36       0.14 5.45        6        11
## px_52          0.05       0.14       -0.14       0.28 3.40        7         6
## px_53         -0.07       0.19       -0.41       0.14 4.76        6         6
## px_54          0.01       0.11       -0.15       0.19 3.73        6        13
## px_55         -0.06       0.15       -0.30       0.09 5.26        6        11
## px_56          0.02       0.16       -0.24       0.39 5.20        6         6
## px_57          0.01       0.26       -0.44       0.51 5.65        6        11
## px_58          0.00       0.19       -0.34       0.29 5.70        6         6
## px_59         -0.00       0.12       -0.12       0.23 4.97        6         6
## px_60          0.08       0.13       -0.13       0.29 4.39        6        16
## px_61         -0.03       0.05       -0.08       0.08 3.13        7         6
## px_62          0.05       0.12       -0.06       0.29 4.58        6        23
## px_63         -0.05       0.16       -0.31       0.13 4.71        6         6
## px_64          0.01       0.13       -0.23       0.24 3.59        7        11
## px_65         -0.10       0.11       -0.35      -0.00 4.98        6         6
## px_66          0.03       0.10       -0.09       0.24 5.86        6        11
## px_67         -0.01       0.15       -0.29       0.32 4.16        6         6
## px_68         -0.05       0.13       -0.30       0.15 4.47        6        11
## px_69          0.07       0.13       -0.11       0.38 5.47        6         6
## px_70         -0.02       0.09       -0.19       0.08 4.70        6        NA
## px_71          0.10       0.11       -0.02       0.31 4.83        6        11
## px_72         -0.04       0.15       -0.36       0.22 4.32        6        11
## px_73         -0.04       0.23       -0.50       0.24 5.12        6         6
## px_74         -0.04       0.12       -0.33       0.13 5.39        6        11
## px_75          0.04       0.05       -0.03       0.12 3.59        7        12
## px_76         -0.07       0.11       -0.29       0.05 5.22        6         6
## px_77         -0.00       0.08       -0.13       0.11 3.83        6         6
## px_78         -0.07       0.10       -0.35       0.03 4.29        6        11
## px_79          0.07       0.12       -0.10       0.28 4.96        6        11
## px_80         -0.00       0.08       -0.12       0.15 5.06        6         6
## px_81          0.06       0.08       -0.05       0.22 4.61        6        11
## px_82         -0.07       0.10       -0.29      -0.00 2.78        7         6
## px_83          0.05       0.17       -0.27       0.38 5.06        6        11
## px_84          0.04       0.11       -0.09       0.28 6.85        6         7
## px_85         -0.06       0.17       -0.35       0.12 4.12        6        11
## px_86         -0.04       0.10       -0.16       0.21 3.85        6        11
## px_87         -0.03       0.14       -0.25       0.19 4.94        6        11
## px_88         -0.01       0.14       -0.18       0.33 5.16        6         6
## px_89          0.06       0.15       -0.18       0.33 6.76        6        NA
## px_90          0.07       0.14       -0.11       0.38 5.90        6         6
## px_91         -0.03       0.08       -0.17       0.09 4.36        6        16
## px_92          0.05       0.15       -0.20       0.25 4.92        7         6
## px_93          0.01       0.13       -0.18       0.26 5.00        6        11
## px_94          0.04       0.13       -0.18       0.33 4.70        6         6
## px_95         -0.04       0.07       -0.15       0.08 4.38        6        11
## px_96         -0.01       0.07       -0.14       0.09 5.80        6        11
## px_97         -0.02       0.06       -0.12       0.09 3.64        7         7
## px_98          0.04       0.08       -0.03       0.20 5.14        6         6
## px_99          0.01       0.11       -0.15       0.30 5.01        6         6
## px_100        -0.11       0.13       -0.34       0.03 4.40        6        NA
## px_101         0.05       0.18       -0.17       0.36 4.51        6         6
## px_102         0.06       0.11       -0.18       0.17 5.24        6        11
```

```
## px_103        -0.01       0.12      -0.19       0.18 4.49        6        11
## px_104         0.03       0.13      -0.15       0.25 5.59        6        11
## px_105        -0.11       0.15      -0.43       0.06 3.15        7         6
## px_106        -0.03       0.08      -0.18       0.08 3.95        6         6
## px_107        -0.01       0.11      -0.39       0.08 3.36        7        11
## px_108        -0.02       0.12      -0.23       0.28 4.72        6         6
## px_109        -0.01       0.07      -0.18       0.06 3.56        7        11
## px_110        -0.05       0.07      -0.25       0.04 4.04        6        11
## px_111         0.05       0.06      -0.05       0.16 4.11        6         6
## px_112        -0.02       0.06      -0.13       0.06 4.95        6         6
## px_113         0.02       0.06      -0.07       0.12 4.09        6        11
## px_114        -0.09       0.14      -0.34       0.05 4.98        6         6
## px_115         0.04       0.16      -0.20       0.34 5.43        6        11
## px_116         0.06       0.16      -0.13       0.40 5.50        6         6
## px_117        -0.00       0.15      -0.31       0.20 5.39        6        11
## px_118        -0.05       0.17      -0.33       0.27 4.32        6         6
## px_119        -0.02       0.17      -0.36       0.15 3.78        7        11
## px_120        -0.02       0.18      -0.31       0.24 5.09        6         6
## px_121         0.08       0.18      -0.16       0.44 5.12        6         6
## px_122        -0.00       0.07      -0.14       0.11 5.83        6         7
## px_123         0.01       0.07      -0.12       0.13 4.65        6         6
## px_124         0.03       0.10      -0.10       0.23 4.97        6        25
## px_125        -0.06       0.11      -0.28       0.07 4.67        6         6
## px_126         0.10       0.09      -0.04       0.26 4.16        6         6
## px_127        -0.09       0.11      -0.26       0.11 4.80        6         6
## px_128         0.04       0.12      -0.13       0.24 5.78        6        11
## px_129        -0.09       0.09      -0.22       0.04 3.35        7         6
## px_130         0.14       0.16      -0.06       0.43 4.04        6         6
## px_131        -0.08       0.19      -0.37       0.22 5.11        6         6
## px_132         0.02       0.21      -0.49       0.24 5.46        6        11
## px_133        -0.05       0.20      -0.25       0.43 6.31        6         6
## px_134         0.01       0.19      -0.44       0.17 6.65        6        11
## px_135         0.07       0.23      -0.20       0.64 4.20        6        11
## px_136        -0.07       0.21      -0.63       0.21 5.50        6        11
## px_137         0.04       0.15      -0.19       0.30 5.14        6         7
## px_138        -0.07       0.16      -0.48       0.17 5.06        6        11
## px_139         0.07       0.14      -0.13       0.32 5.62        6         7
## px_140        -0.08       0.17      -0.43       0.08 5.04        6         6
## px_141         0.12       0.15      -0.02       0.44 5.71        6         7
## px_142        -0.16       0.12      -0.40      -0.05 6.50        6        NA
## px_143         0.10       0.17      -0.26       0.32 5.42        6        11
## px_144        -0.04       0.13      -0.22       0.16 5.30        6         6
## px_145         0.07       0.07      -0.04       0.15 4.27        6         6
## px_146        -0.08       0.08      -0.18       0.06 4.33        6        21
## px_147        -0.02       0.15      -0.37       0.13 6.53        6        11
## px_148         0.06       0.22      -0.17       0.71 6.14        6         6
## px_149         0.04       0.21      -0.55       0.23 6.05        6        11
## px_150        -0.02       0.17      -0.16       0.41 5.71        6         7
## px_151        -0.03       0.21      -0.54       0.21 4.77        6        11
## px_152         0.00       0.22      -0.25       0.48 5.82        6         7
## px_153         0.03       0.20      -0.33       0.29 6.65        6        11
## px_154        -0.02       0.21      -0.31       0.51 5.86        6         7
## px_155         0.04       0.12      -0.10       0.20 7.06        6        11
## px_156         0.02       0.09      -0.14       0.18 5.18        6         7
```

```
## px_157        -0.11       0.09      -0.28       0.03 3.47        7        11
## px_158         0.14       0.09       0.01       0.32 3.77        6         7
## px_159        -0.07       0.11      -0.19       0.21 5.00        6         6
## px_160        -0.02       0.07      -0.18       0.08 5.79        6        NA
## px_161        -0.10       0.11      -0.30       0.05 5.46        6         6
## px_162         0.14       0.15      -0.07       0.37 5.10        6         6
## px_163        -0.08       0.12      -0.26       0.10 4.77        6         6
## px_164         0.01       0.12      -0.31       0.11 5.33        6        11
## px_165        -0.12       0.13      -0.32       0.06 7.06        6         6
## px_166         0.13       0.11      -0.08       0.32 3.84        6        11
## px_167        -0.07       0.14      -0.32       0.11 5.12        6         6
## px_168         0.08       0.11      -0.08       0.26 4.79        6        11
## px_169        -0.07       0.12      -0.32       0.08 4.26        6         6
## px_170         0.06       0.15      -0.14       0.36 5.75        6        11
## px_171        -0.05       0.13      -0.28       0.08 4.34        6         6
## px_172        -0.03       0.08      -0.17       0.14 6.21        6        11
## px_173         0.11       0.08      -0.02       0.27 5.04        6        11
## px_174        -0.10       0.06      -0.21      -0.00 3.48        7        15
## px_175         0.07       0.10      -0.10       0.24 5.35        6        11
## px_176        -0.00       0.04      -0.05       0.08 3.82        7         7
## px_177         0.03       0.11      -0.19       0.16 5.46        6        11
## px_178        -0.03       0.14      -0.24       0.24 4.37        6        11
## px_179        -0.02       0.10      -0.20       0.13 4.49        6         6
## px_180         0.03       0.08      -0.08       0.17 4.82        6         7
## px_181         0.08       0.05       0.01       0.19 2.88        7         7
## px_182        -0.09       0.08      -0.27       0.03 3.66        6        11
## px_183         0.05       0.07      -0.05       0.18 5.73        6         6
## px_184        -0.07       0.11      -0.26       0.15 3.69        6        11
## px_185         0.05       0.06      -0.07       0.14 4.39        6         7
## px_186        -0.05       0.09      -0.23       0.06 4.93        6         6
## px_187         0.06       0.13      -0.08       0.38 3.42        7        11
## px_188        -0.02       0.16      -0.26       0.25 5.92        6        11
## px_189        -0.08       0.12      -0.27       0.07 5.23        6         6
## px_190         0.03       0.06      -0.09       0.10 5.04        6         6
## px_191        -0.01       0.10      -0.17       0.18 6.23        6         6
## px_192        -0.05       0.06      -0.15       0.05 4.67        6         6
## px_193        -0.05       0.15      -0.30       0.12 4.63        6         6
## px_194         0.02       0.16      -0.20       0.24 3.58        7        11
## px_195         0.01       0.10      -0.17       0.19 6.91        6        NA
## px_196        -0.02       0.12      -0.24       0.14 5.46        6         6
## px_197        -0.05       0.05      -0.17       0.01 5.10        7         7
## px_198         0.08       0.09      -0.06       0.22 5.45        6        11
## px_199        -0.07       0.14      -0.28       0.21 5.20        6        11
## px_200         0.01       0.11      -0.10       0.22 5.87        6         7
## px_201        -0.01       0.07      -0.11       0.15 4.02        6        11
## px_202         0.05       0.11      -0.18       0.22 5.03        6        11
## px_203        -0.09       0.13      -0.33       0.07 3.77        6        11
## px_204         0.05       0.09      -0.10       0.21 5.31        6        11
## px_205         0.06       0.09      -0.04       0.27 5.85        6        11
## px_206         0.01       0.15      -0.16       0.31 5.38        6        11
## px_207        -0.05       0.08      -0.21       0.04 4.52        6        11
## px_208         0.09       0.11      -0.03       0.30 4.59        6         6
## px_209         0.02       0.06      -0.06       0.13 4.29        6        11
## px_210        -0.04       0.08      -0.20       0.09 4.57        6        14
```

```
## px_211          0.01       0.10      -0.08       0.24 5.32        6        6
## px_212          0.00       0.13      -0.23       0.21 5.24        6       11
## px_213          0.04       0.08      -0.10       0.18 5.62        6        6
## px_214         -0.06       0.10      -0.22       0.11 5.48        6       11
## px_215          0.09       0.13      -0.13       0.26 5.27        6        6
## px_216         -0.01       0.13      -0.17       0.23 5.18        6       11
## px_217         -0.04       0.07      -0.17       0.08 5.84        6        7
## px_218         -0.00       0.12      -0.18       0.18 6.11        6        6
## px_219          0.09       0.21      -0.18       0.48 5.54        6        7
## px_220         -0.11       0.21      -0.54       0.07 4.38        6        6
## px_221          0.02       0.23      -0.21       0.51 4.90        6       12
## px_222         -0.06       0.27      -0.65       0.14 4.66        6        6
## px_223          0.06       0.14      -0.16       0.32 6.33        6        6
## px_224         -0.11       0.15      -0.41       0.05 5.17        6        6
## px_225         -0.02       0.13      -0.29       0.12 7.00        6        6
## px_226          0.07       0.19      -0.15       0.46 4.72        6        6
## px_227         -0.06       0.18      -0.39       0.16 4.66        6        6
## px_228          0.05       0.23      -0.20       0.49 6.63        6        6
## px_229         -0.08       0.21      -0.48       0.15 5.29        6        6
## px_230          0.04       0.22      -0.21       0.44 5.26        6        7
## px_231         -0.05       0.20      -0.38       0.16 5.69        6        6
## px_232          0.03       0.17      -0.19       0.29 5.14        6       17
## px_233          0.07       0.10      -0.11       0.23 4.99        6       11
## px_234         -0.04       0.12      -0.30       0.07 6.87        6       NA
## px_235         -0.04       0.20      -0.41       0.29 6.68        6        6
## px_236          0.07       0.22      -0.21       0.51 5.42        6        7
## px_237          0.01       0.19      -0.38       0.22 5.10        6        6
## px_238          0.02       0.22      -0.17       0.50 3.84        6       23
## px_239          0.02       0.09      -0.13       0.14 6.08        6        6
## px_240          0.05       0.09      -0.09       0.21 5.56        6       NA
## px_241          0.00       0.14      -0.16       0.28 4.37        6       11
## px_242         -0.05       0.18      -0.41       0.15 3.82        6        6
## px_243          0.02       0.13      -0.20       0.25 4.12        6       11
## px_244         -0.04       0.16      -0.37       0.11 7.52        6        6
## px_245          0.05       0.18      -0.14       0.40 5.47        6        7
## px_246         -0.06       0.22      -0.45       0.20 5.38        6        6
## px_247          0.00       0.15      -0.19       0.23 3.93        6       11
## px_248          0.01       0.13      -0.16       0.20 3.97        6        7
## px_249         -0.06       0.08      -0.24       0.07 4.96        6        7
## px_250         -0.02       0.14      -0.20       0.30 5.52        6       11
## px_251          0.05       0.20      -0.32       0.41 4.50        6        7
## px_252         -0.09       0.24      -0.50       0.23 3.95        6        7
## px_253          0.06       0.27      -0.18       0.53 4.54        6       22
## px_254         -0.12       0.30      -0.68       0.17 4.84        6        6
## px_255          0.09       0.21      -0.13       0.46 6.51        6        6
## px_256         -0.13       0.15      -0.42       0.02 6.12        6        6
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
post_draws <- as.data.frame(brm_logit)
```

Model validation

```
library(caret)
```

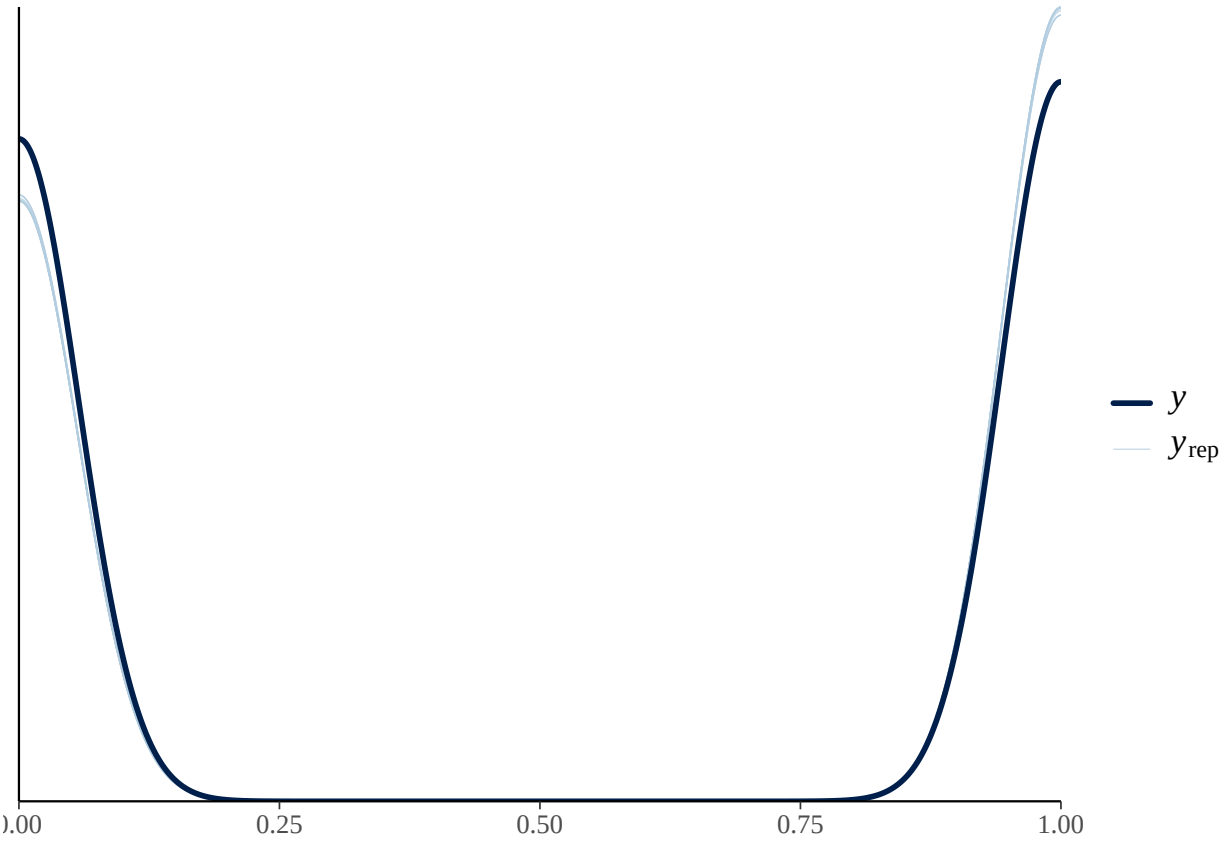
```
## Loading required package: lattice
```

```
##
## Attaching package: 'caret'
```

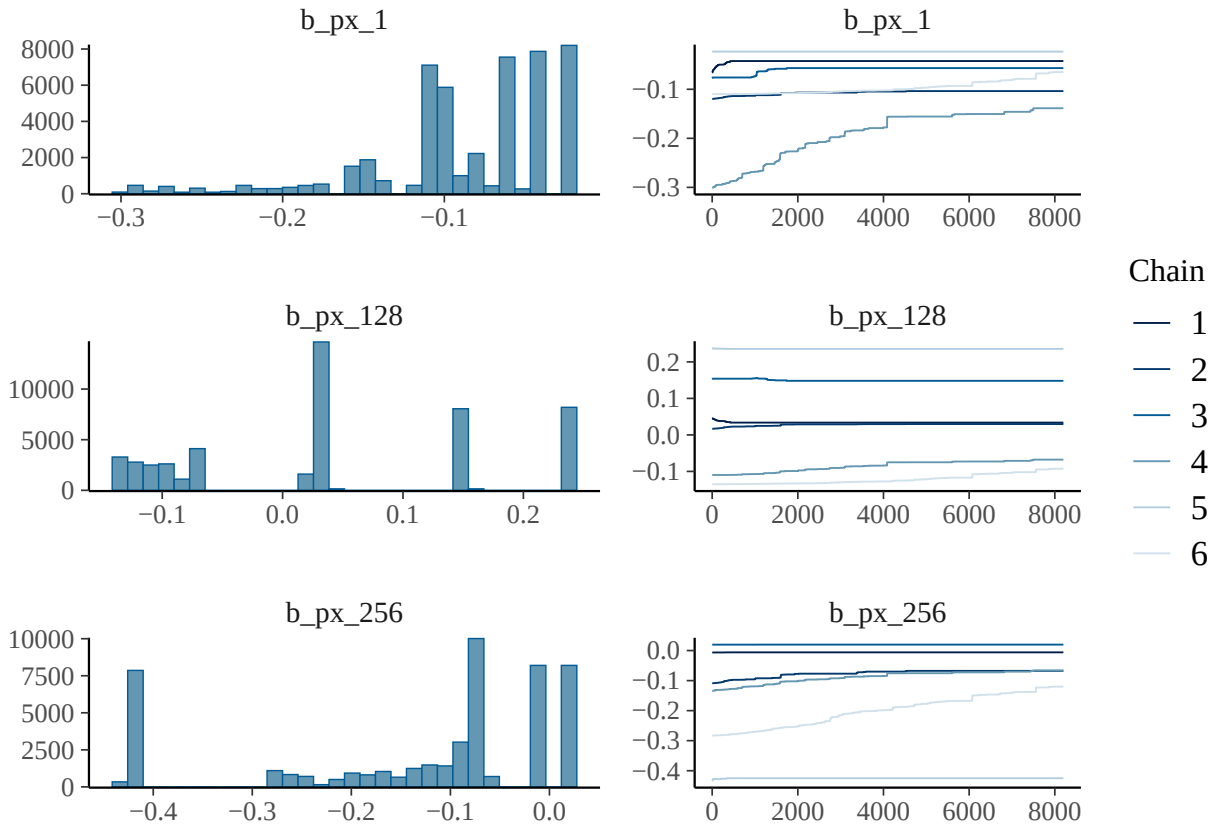
```
## The following object is masked from 'package:purrr':
##
##     lift
```

```
# Check posterior predictive fit
pp_check(brm_logit)
```

```
## Using 10 posterior draws for ppc type 'dens_overlay' by default.
```



```r
# Plot individual features
plot(brm_logit, variable = c("b_px_1", "b_px_128", "b_px_256"))
```




```r
# Evaluate on validation set
pred_probs <- posterior_epred(brm_logit, newdata = valid_df)
pred_label <- as.numeric(colMeans(pred_probs) > 0.5)

# Confusion matrix
confusionMatrix(factor(pred_label, levels=c(1,0)),
                factor(valid_df$label, levels=c(1,0)),
                mode = "everything")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    1    0
##          1 2903  520
##          0  577 2300
##
##                Accuracy : 0.8259
##                  95% CI : (0.8163, 0.8352)
##     No Information Rate : 0.5524
##     P-Value [Acc > NIR] : < 2e-16
##
##                   Kappa : 0.6486
```

```
##
##  Mcnemar's Test P-Value : 0.09088
##
##             Sensitivity : 0.8342
##             Specificity : 0.8156
##          Pos Pred Value : 0.8481
##          Neg Pred Value : 0.7994
##               Precision : 0.8481
##                  Recall : 0.8342
##                      F1 : 0.8411
##              Prevalence : 0.5524
##          Detection Rate : 0.4608
##    Detection Prevalence : 0.5433
##       Balanced Accuracy : 0.8249
##
##        'Positive' Class : 1
##
```

Model testing

```
# Evaluate on test set
pred_probs <- posterior_epred(brm_logit, newdata = test_df)
pred_label <- as.numeric(colMeans(pred_probs) > 0.5)

# Confusion matrix, get F1, precision, recall
test_conf_matrix <- confusionMatrix(factor(pred_label, levels=c(1,0)),
                factor(test_df$label, levels=c(1,0)),
                mode = "everything")
test_conf_matrix
```

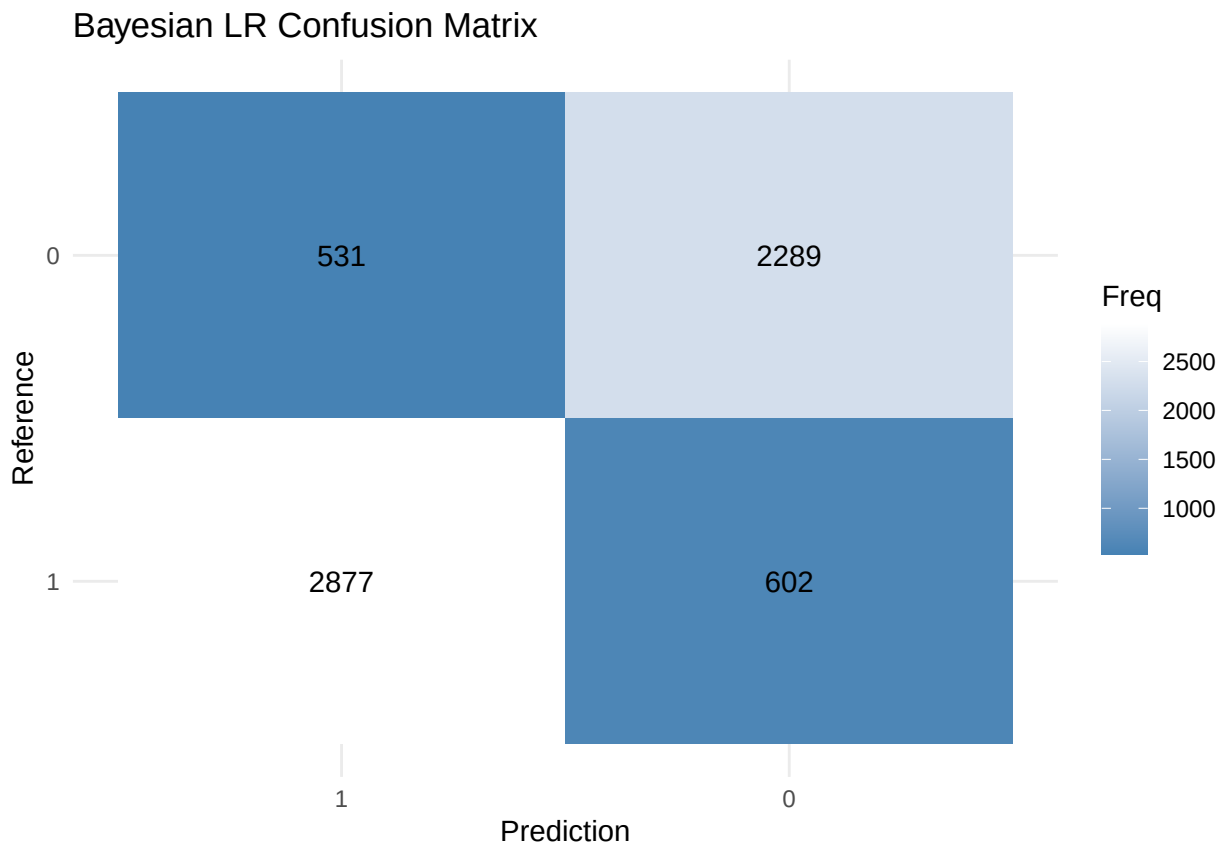
```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    1    0
##          1 2877  531
##          0  602 2289
##
##                Accuracy : 0.8201
##                  95% CI : (0.8104, 0.8295)
##     No Information Rate : 0.5523
##     P-Value [Acc > NIR] : < 2e-16
##
##                   Kappa : 0.6371
##
##  Mcnemar's Test P-Value : 0.03756
##
##             Sensitivity : 0.8270
##             Specificity : 0.8117
##          Pos Pred Value : 0.8442
##          Neg Pred Value : 0.7918
##               Precision : 0.8442
##                  Recall : 0.8270
```

```
##                      F1 : 0.8355
##              Prevalence : 0.5523
##          Detection Rate : 0.4567
##    Detection Prevalence : 0.5410
##       Balanced Accuracy : 0.8193
##
##        'Positive' Class : 1
##
```

```r
# Get per-class accuracies
test_conf_mat_2 <- table(Predicted = pred_label, Actual = test_df$label)
true_positives <- diag(test_conf_mat_2) # TP
actual_counts <- colSums(test_conf_mat_2) # Total per class
per_class_accuracy <- true_positives / actual_counts # Per-class acc
print(per_class_accuracy)
```

```
##         0         1
## 0.8117021 0.8269618
```

```r
# Generate confusion matrix figure
library(ggplot2)
plt <- as.data.frame(test_conf_matrix$table)
ggplot(plt, aes(Prediction, Reference, fill = Freq)) +
  geom_tile() +
  geom_text(aes(label = Freq)) +
  scale_fill_gradient(low = "steelblue", high = "white") +
  ggtitle("Bayesian LR Confusion Matrix") +
  theme_minimal()
```


Bayesian LR Confusion Matrix
Reference
0
1
531
2289
2877
602
1
0
Prediction
Freq
2500
2000
1500
1000